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An Initial Investigation of Tungsten Inert Gas (TIG) Torch as Heat Source for Additive Manufacturing (AM) Process

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Abstract

Additive Manufacturing (AM) processes are gaining more steam in the last years, due to a series of advantages, such as capability of producing complex parts never possible before, low stock to be removed, complex parts in exotic materials, etc. However, energy efficiency, high costs and longer fabrication time, could be some of the drawbacks in the future. Welding processes can be used to produce pieces by AM techniques and their application has been intensified lately. Commonly, AM processes use MIG/MG and TIG and primarily with wire as stock material. Welding offers the advantage of producing large parts in lower time and material consistency is always good, since those processes have been greatly improved. In contrast, welding workpieces, generally, result with rough surfaces and poor geometric quality, which makes them not readily acceptable for most of the common machinery applications. The present work investigates the use TIG process with metal powder as stock material to produce workpieces, which are to be post processed by machining. The AM process is based on the Powder Bed Fusion (PBF) system. A simple one-axis moving system was produced to test the whole concept. Powder was preferred since it is readily available in variety of metallic alloys. Two powder grain sizes (*Gs*), three travel speed values (*f*) and three current levels (*C*) were tested producing weld beads of AISI H13 on a substrate of AISI 1020 plate. Beads were assessed by measuring external dimensions and looking at their microstructures. Additional tests were performed building straight wall stacking 10 layers with the best welding parameters found with the first trials. Results indicate that materials with good internal quality could be produced when making single beads and also building straight walls. These first trials show that the proposed process can be used as a promising hybrid process, using AM (PBF-TIG) and conventional machining at the same equipment. Future work will concentrate on adapting the welding and powder bed system in a machining center to further study the process.

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1. Introduction

For the last years, Additive Manufacturing (AM) processes have been intensively adopted by the most different fields, such as, aeronautical, military, medical, etc. and producing some encouraging results to manufacture complex and high-value components. In general, 3D parts can be designed in Computer Aided Design (CAD) software, and then produced, without the need of conventional tooling. The use of high strength and

reactive materials are one of the most attractive aspects of this new technology. According to ISO/ASTM2900-15 there are 7 categories of AM processes and two of them – Powder bed Fusion (PBF) and Direct Energy Deposition (DED) – use a beam (laser or electron) or an electric arc as primary energy source for melting the metal layer by layer through the deposition process.

The use of laser and electron beam is more advanced and specific machines for such applications are already available in

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the market. Arc welding equipment, however, is available, but just few specifically for AM applications. Among the vast variety of equipment and welding processes, Tungsten Inert Gas (TIG), Metal Inert Gas (MIG) and plasma are progressively being used in attempts to build 3D shapes in the way DED processes do. Wire Arc Additive Manufacturing (WAAM) offers high deposition rates, good quality material without size limits, becoming the best alternative for AM of medium to large size parts with high mechanical requirements. Twenty-century improvements in arc welding include the use of a continuous wire feed and process automation using robots. Shapes built are not dimensionally accurate as those obtained by PBF or laser DED, for example, but that can be overcome by encompassing subtractive processes, such as machining, in hybrid machines such as ROMI D800 Hybrid.

Metal Inert/Active Gas (MIG/MAG) establishes an electric arc between a consumable wire electrode and the substrate producing a melt pool shielded by a gas that be inert (He or Ar) or active (mostly CO₂ or other gas mixtures) [1]. Melting and mixing both materials in the pool promote the addition. Tungsten Inert Gas (TIG) process, in which a non-consumable tungsten rod is used to initiate the electric arc under the protection of an inert gas, is commonly used to join a wide variety of materials including steels, Ni-based, Ti-based and Aluminum alloys, for example. It is a more sensitive process and its use is more restricted to manual welding using wire when material need to be added [2].

Tabereroa et. al. [3] tested several welding processes for 3D deposition shapes: Plasma Welding (PAW), Cold Metal Transfer (CMT) and TIG with “co-axial” wire feed system. A Ti alloy and stainless steel were deposited and TIG process could be used for small/medium size Titanium and Stainless Steel (SS) parts, which have medium/high mechanical requirements. Results show that the material exhibits an almost isotropic feature in tensile properties and very low porosities. Such result indicates the possibility of using TIG heating source for AM processes. On another research, Rodriguez et. al. [4] compared the applicability of two arc welding technologies, cold metal transfer (CMT) and TIG for AM in stainless steel parts with continuous and pulsed current. Large shapes, such as, straight walls and square forms were built in bead-by-bead strategy. Most of the material produced, in that work exhibited certain anisotropy in mechanical properties in the as-built state with lower strength values in the vertical (Z) orientation, which is the stacking direction.

Several problems of deposition accuracy result during Gas Tungsten Arc Welding (GTAW) based additive manufacturing when wire is fed through side direction. Genga et. al. [5] addressed such problems using, high speed camera and mathematical models. Changes in wire entering angle (10–20°) can significantly modify the metal melting and transferring mechanism, affecting the bead appearance and accuracy. Hejrjipour et. al. [6] investigated heat transfer, fluid flow and mass transport using numerical simulation and performed experiments to verify the numerical results in a first clad layer profile, (width and height). Seventy per cent of the clad layer is made up of the wire filler only about 30% of the substrate mixed into a relatively homogeneous clad layer.

Most of the application with electric arc as heat source uses wire as feedstock, although powder could be an alternative. Plasma transfer arc (PTA) is a common process for cladding in metals and use powder more frequently, although not yet used to build 3D parts. Cardozo et. al. [7] tested and compared PTA with wire and powder as feedstock material obtained from Inconel 625. It was found that productivity results higher and the microstructure coarser using wire. Powder, however, has a higher solidification rate, which accounts for residual stresses at the top of the deposits. Post-deposition heat treatment was also used and reduced the differences due to feedstock form used and caused the precipitation of δ phase at the interface with the substrate.

Bond et. al. [8] investigated the role of the atomized feedstock on the solidification and powder grain size aspects as a processing variable using PTA. A model containing five separated temperature conditions, by which a metallic part can passes through during its interaction with the plasma, is proposed. That temperature is a function of deposition electrical current and the particle size. Hardness is predominantly affected by such temperature. In view of all that, arc welding seems to be a very promising technology for AM in the same way that metallic powder is a suitable feedstock material to achieve good material quality. Although parts produced by welding may not result in readily useful ones, post processing by machining, in a hybrid process, remains as an alternative.

The present work proposes an innovative AM process using TIG arc welding as heating source, metal powder as feedstock material in a PBF physical arrangement (PBF-TIG). Initials tests were performed to validate the process, which considered the production of single bead straight lines using several process parameters and two powder grain sizes of AISI H13 tool steel. After this trial, two walls stacking layer-by-layer were also deposited with the best two set of parameters.

2. Methodology

To prove the concept, a simple experimental set up was put together. The built machinery combines a continuous movement in one axis that carries the both the powder system and the torch at the same defined travel speed. The powder delivering system has similar concept to a powder bed fusion (PBF) system. A TIG torch melts a bed of powder, which after solidifying, produces single beads of about 30–50 mm length. Figure 1 shows an overall view of the set up and selected details.

The torch used were the straight model, normally used in robots, and the thorium-tungsten electrode of 2.4 mm diameter with 4 mm left out of the gas lens, 4 mm from bed of powder, which has 1.5 mm thickness. That thickness was guaranteed by a wiper. Figure 2 shows more details of the TIG torch. The powder duct, combined with the wiper, is capable of spreading a flat and uniform layer of 1.5 mm thick bed of powder, as shown in Figure 3.

Initial experiments were planned to explore the feasibility and limitations of the proposed process. Powder grain size (*Gs*) of AISI H13, travelling speed (*f*) and current (*C*) were the parameters tested to produce a single weld bead. The beads

were deposited on a 3 mm thick AISI 1020 carbon steel around 25 x 60 mm. Gas flow of 7 L/min and torch angle of 90° were used at all trials.

Table 1 shows the working conditions for these first experiments.

To further test the proposed system, and the setup, trials were performed to build straight walls. Using the best parameter set found in the first trials, two walls were built using the two different grain sizes (*Gs*). Ten layers were stacked leaving about 30 s of dwelling time between them.

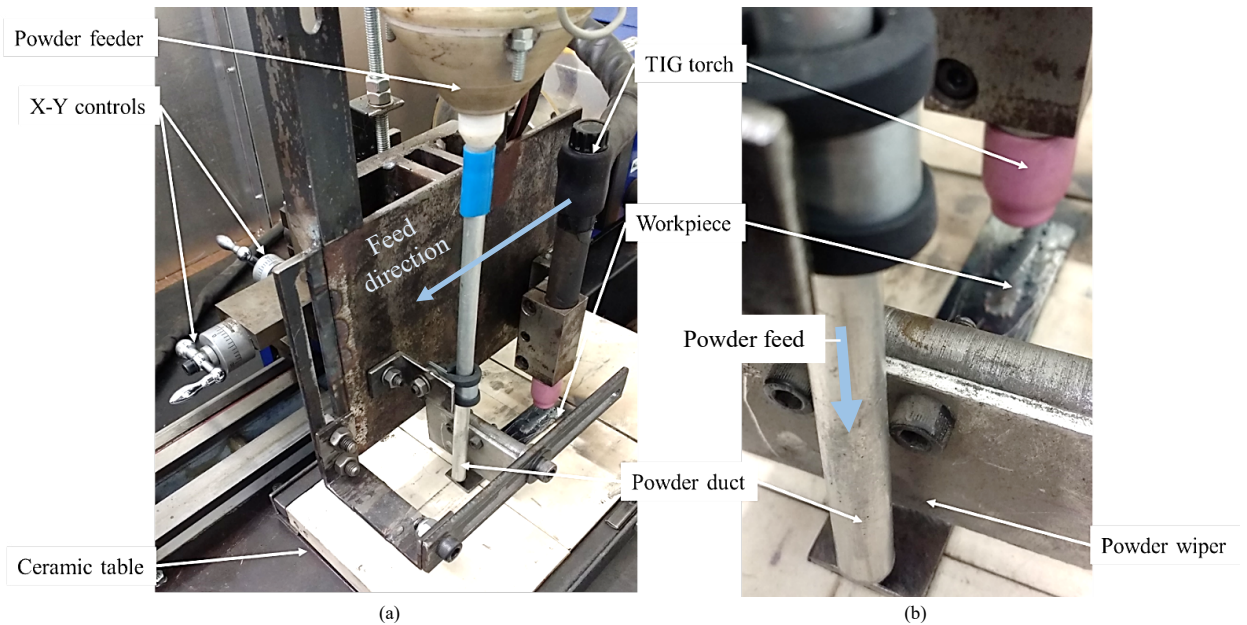


Fig. 1. Overall view of the set up used to test the feasibility of the system used to produce AM parts using TIG welding torch. (a) overall view, (b) torch and workpiece detail.

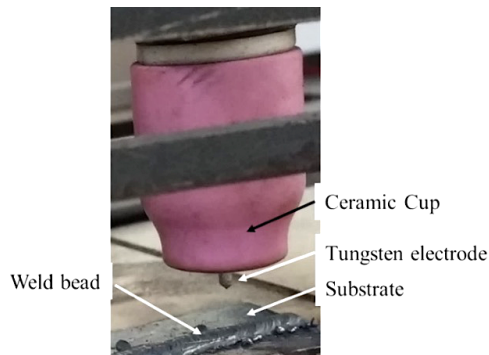


Fig. 2. Details of the TIG torch used in the experiments.

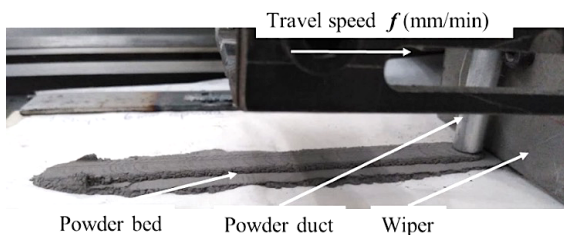


Fig. 3. Details of the powder bed being formed by the powder duct and wiper.

Table 1. Working conditions for the first experiments exploring the feasibility and limitations of TIG-PBF AM process.

Trial	f (mm/min)	Gs (μm)	C (A)
1	37.5	53-105	50
2	47.5	53-105	50
3	56.5	53-105	50
4	37.5	53-105	70
5	47.5	53-105	70
6	56.5	53-105	70
7	37.5	53-105	60
8	47.5	53-105	60
9	56.5	15-45	60
10	37.5	15-45	50
11	47.5	15-45	50
12	56.5	15-45	50
13	37.5	15-45	70
14	47.5	15-45	70
15	56.5	15-45	70
16	37.5	15-45	60
17	47.5	15-45	60
18	56.5	15-45	60

The powder was spread and immediately fused by the torch through the whole length of about 50-70 mm. Then, the deposition head returned to the initial position, and a new layer started in the same way.

Cross sections from some single beads and from the two walls were cut, polished and etched with Nital 2%, to examine the external geometry. To evaluate the quality of the deposited material, a Buchler microhardness tester (model 1600-6300) was used to measure microhardness Vickers (HV) through the cross section of the built wall with the load of 1 kgf.

3. Results and discussion

Figure 4 shows the typical external aspect of single beads obtained with the initial setup.

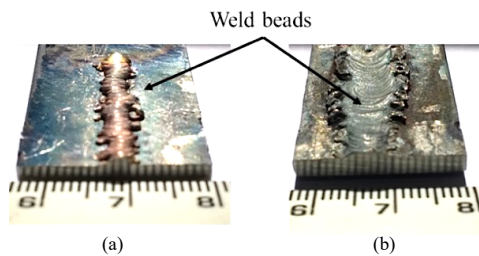


Fig. 4. Typical aspects of the single beads obtained with the PBF-TIG systems. (a) $f = 47.5$ mm/min, $G_s = 53-105$ μm and $C = 50$ A, (b) $f = 37.5$ mm/min, $G_s = 15-45$ μm and $C = 60$ A.

It can be noticed that the external aspects are typical of TIG welding beads obtained with powder as feedstock [8].

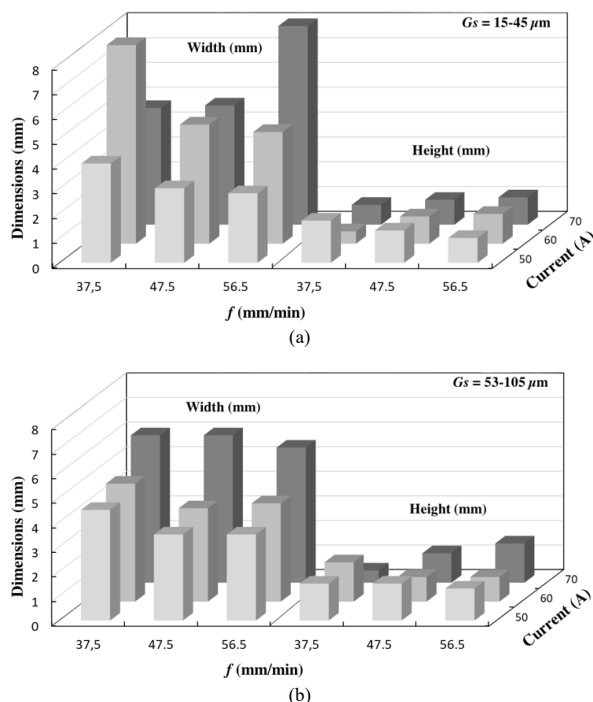


Fig. 5. Overall dimensions of beads obtained with the PBF-TIG process. (a) $G_s = 53-105$ μm (b) $G_s = 15-45$ μm .

There are many balling features on the bead sides, as a consequence of low power at those points, which was not capable of melting the substrate. When the power is high enough to melt the powder on the bed, but not the substrate, the molten metal assumes a spherical shape before solidifying. External dimensions of the weld beads obtained are shown in Figure 5.

In general, the beads were from 3.0 to 8.0 mm wide and from 0.5 to 1.7 mm high. Current (C) and travel speed (f) affected the external dimensions of single beads, and to a less extent, the powder grain size too. Higher values of current combined with lower travel speed produced shorter and wider beads. The interaction between parameters in the formation of the melt pool is complex. In this perspective, it has been seen that different combinations of current and travel speed can produce beads with the same dimensions with both powder-grain sizes, as well as at certain levels, one parameter can stand over the others and change the outcome.

For example, the sets in which current is 50 and 60 A the overall height increases with the rise of travel speed. However, for the sets that ran with 70 A, it is shown otherwise. This can be explained due to the higher energy is transferred to the powder/substrate. Coarser powder particles can take advantage in heat transfer by conduction, whereas finer powders might have a higher rate of disintegration with such high energy delivered at the melt pool. In this case, the melt pool runs out of material, as the powder is not fed coaxially, resulting in a lower height of the bed – when compared to larger particle size powder at the same conditions.

In this perspective, widest beads have been produced when $C = 60$ A, $f = 37.5$ mm/min and $G_s = 15-45$ μm . As aforementioned, the higher the current applied, the higher the energy transferred to the powder/substrate. Coarser powder particles can distribute the heat transfer proportionally to height-width dimensions, whereas finer powders can disintegrate with such high energy delivered at the melt pool, weakening the fusion in height mainly. In this case, as the powder is not fed coaxially, the placement of melt pool and its geometry can be causing the production of wider beads by finer powder.

When the cross section of individual beads was cut, the typical aspect obtained is shown in Figure 6.

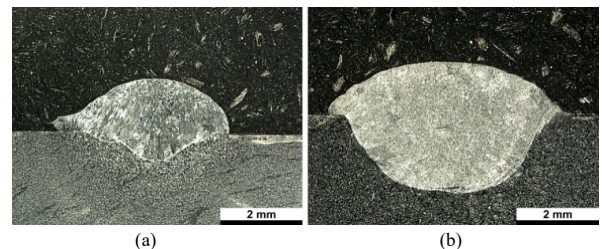


Fig. 6. Typical cross section (etched with Nital 2%) of a single bead obtained with PBF-TIG process. (a) $f = 47.5$ mm/min, $G_s = 53-105$ μm and $C = 50$ A, (b) $f = 37.5$ mm/min, $G_s = 15-45$ μm and $C = 60$ A.

Figure 6 shows that depending on the parameters used the shape of cross section and the depth of penetration on the substrate result differently. The values of current affect

significantly the depth of deposition on the substrate and that can be used with an advantage in favor of AM process. If the objective is to build a piece that must be separated from the substrate after building, low current should be used, and the adhesion will be lower than with higher current. On the other hand, if the objective is to repair and a good adhesion is needed current level can be increased to improve such aspect.

Figure 7 shows a general aspect of the two walls built with the best welding conditions for each of the powder grain size.

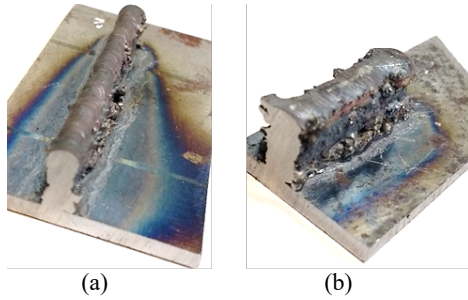


Fig. 7. External aspects of the walls built with PBF-TIG process in (a) perspective view and (b) side view. (a) $f = 37.5$ mm/min, $G_s = 53-105$ μm and $C = 60$ A, (b) $f = 47.5$ mm/min, $G_s = 15-45$ μm and $C = 60$ A.

In general, the surface finishing and external aspects of the walls built with different powder grain size resulted rough with loose dimension control, therefore machining must be performed as a post processes to make some kind of useful part. Many balling features were also obtained reminding satellites particles adhered on the lateral surfaces of both walls.

Table 2 summarizes the measures of the external dimensions of the obtained walls taken at every 5 mm throughout the extension of the workpiece.

Table 2. External dimensions of the two walls obtained the two different Powder grain sizes.

G_s (μm)	Average height (mm)	Average Width (mm)
15-45	11.88 ± 0.17	6.63 ± 0.94
53-105	11.27 ± 0.12	5.25 ± 0.42

In average the walls produced with finer grains resulted slightly taller and wider (11.88×6.63 mm) than that produced with coarser powder (11.27×5.25 mm). The wider wall can be related to the heat abortion, which is probably higher with the finer powder, as the particles are find in a more compact way, leading to a wider area of heat transfer, leading to the fusion of material when the torch ran over the powder bed. Similar results were reported by Bond et. al. [8] when increasing the percentage of finer powder. Finer grains tend to reach higher temperatures when passing through the plasma of the TIG torch.

The height of the walls seems to be related with the higher compaction of the finer powder compared to the coarser one. The major difference that can explain these geometry differences is that in a coarser material, more vacancies is found within the particles of powder in the bed. In this case, majority of the heat is transferred by conduction in a particle-

particle interaction, and some by convection in a particle-vacancy boundary. during the torch movement.

Figure 8 shows the cross section of the walls obtained with the best conditions for each of the powder sizes. The cross sections of the wall show a very rough profile, one aspect which has to be improved as such process evolves.

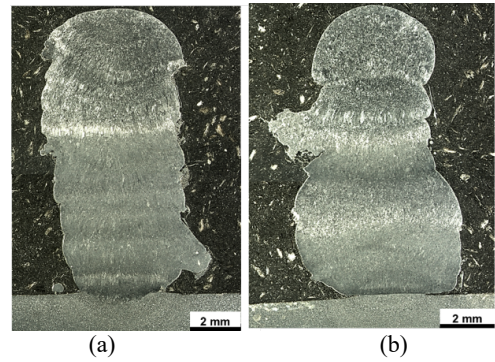


Fig. 8. Cross sections of the walls obtained with PBF-TIG using (a) coarse powder and (b) fine powder.

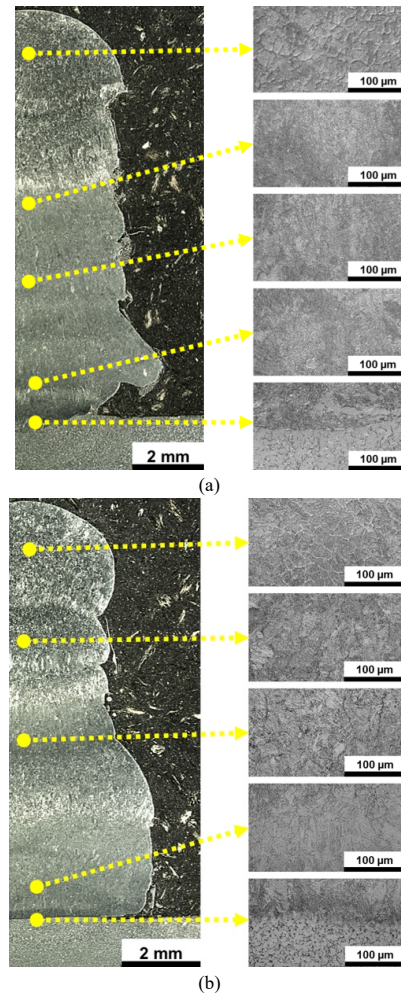


Fig. 9. Typical microstructures found at several points of the cross section of the walls, (a) coarse powder and (b) fine powder.

The moving system hereby built was mainly to find the best welding parameter for the TIG equipment. Now that TIG parameters have been tested and proved the torch will be moved to a machining center, in which all geometric and kinematic problems will not be an issue anymore. Therefore, better external aspects of the future walls must be achieved.

Figure 9 shows some aspects of the microstructures found at several points of the walls. Microstructures were typical of AISI H13 and there was good fusion amongst layers and lateral waviness were not very distinct from those typically obtained with AM using welding techniques [2, 4].

Grains are relatively large, and some martensite could be found in some parts of the wall, especially at higher points. There was no clear distinction between layers, which seems to indicate that melting was complete during deposition.

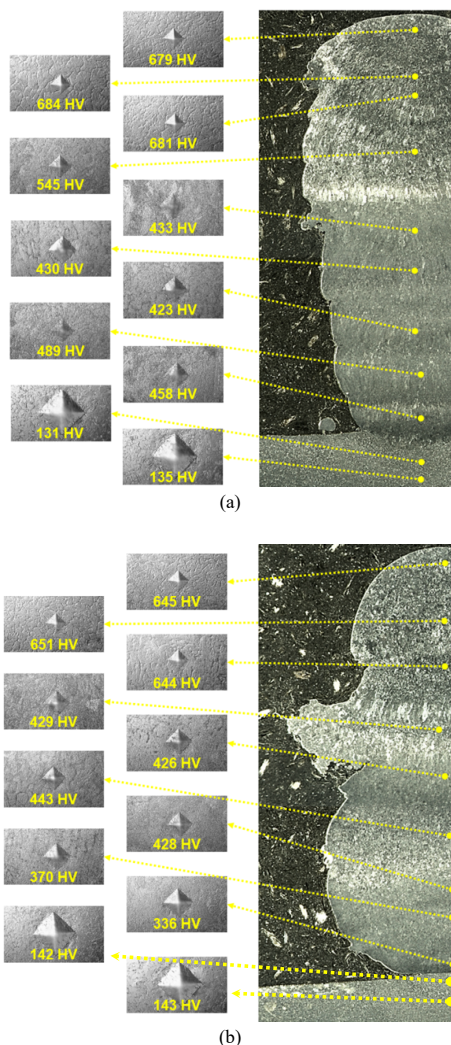


Fig. 10. Hardness values along the height of the walls, (a) coarse powder and (b) fine powder.

Figure 10 shows micro hardness values along the height of the cross sectioned walls. After analysis, the wall made with coarser powder was machined to the final shape of a regular straight and slender wall. Figure 11 shows the result.



Fig. 11. External aspect of a wall after being machined.

The general aspect of the machined wall is similar to an equivalent workpiece fully machined from a solid block, except that this particular one contains two different materials and less steel was transformed into swarf. Machining time was far less than starting with a conventional solid blank. Overall, the whole process of building a generic AISI H13 workpiece using TIG-PBF process with powder was demonstrated. Future work will include setting the TIG torch on a machining center with a proper powder bed system to explore the full capability of such process to produce more complex shapes, therefore enabling the performance of further mechanical tests and analysis. It is possible to foresee some interesting applications for such process, since it tends to encompass some qualities of the PBF system, i.e. layer-by-layer building capacity, with some from the TIG welding, i.e. low cost, high energy efficiency, good material quality and low operation cost, at the same process. Much has yet to be done to improve geometry and surface quality, in order to become a useful AM process.

4. Conclusions

From these first experiments performed to test the TIG-PBF process, one can reach the following conclusion:

- The use of a TIG torch as heating source to produce Additive Manufacturing (AM) pieces demonstrates to be feasible in a system similar to a Powder Bed Fusion (PBF) system;
- Weld beads could be produced using a very simple mechanism moving in straight line. Cross section dimensions were very dependent on a combination of traveling speed and electrical current. Combinations of low traveling speed and high current tend to produce wider and shorter cross section;
- Using a very simple moving mechanism straight walls were produced with relatively good microstructure and good adhesion amongst layers. Hardness was lower at the bottom of the walls, but between 336 and 684 HV, depending on parameters combination;
- After machined, one wall of 3.0 mm wide and 10 mm height was produced with good quality for mechanical applications;
- Although some simple pieces could be produced using the one-axis moving system hereby produced, much has to be done to improve piece geometry and surface quality. Using

a CNC machine, capable of producing complex parts by AM and post process them by machining, seems to be the way towards an innovative hybrid process.

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References

- [1] Lordachescu D, Quintino L, Miranda R, Pimenta G. Influence of shielding gases and process parameters on metal transfer and bead shape in MIG brazed joints of the thin zinc coated steel plates. *Materials & Design* 2006;27:381-390.
- [2] Bai JY, Yang CL, Lin SB, Dong BL, Fan CL. Mechanical properties of 2219-Al components produced by additive manufacturing with TIG. *The International Journal of Advanced Manufacturing Technology* 2016;86:479-485.
- [3] Tabernero I, Paskual A, Álvarez P, Suárez A. Study on Arc Welding processes for High Deposition Rate Additive Manufacturing. *Procedia CIRP* 2018;68:358-362.
- [4] Rodríguez N, Vázquez L. Wire and arc additive manufacturing: a comparison between CMT and TopTIG processes applied to stainless steel. *Welding in the World* 2018;62:1083-1096.
- [5] Geng H, Li J, Xiong J, Lin X, Zhang F. Optimization of wire feed for GTAW based additive manufacturing. *Journal of Materials Processing Technology* 2017;243:40-47.
- [6] Hejripour F, Valentine DT, Aidun DK. Study of mass transport in cold wire deposition for Wire Arc Additive Manufacturing. *International Journal of Heat and Mass Transfer* 2018;125:471-484.
- [7] Cardozo EP, Rios S, Ganguly S, D'Oliveira ASCM. Assessment of the effect of different forms of Inconel 625 alloy feedstock in Plasma Transferred Arc (PTA) additive manufacturing. *The International Journal of Advanced Manufacturing Technology* 2018;98:1695-1705.
- [8] Bond D, D'Oliveira ASCM. Effect of Current and Atomized Grain Size Distribution on the Solidification of Plasma Transferred Arc Coatings. *Materials Research* 2012;15:770-774.
- [9] ASM Handbook: volume 4A: Steel Heat Treating Fundamentals and Processes. ASM International 2013; ISBN-10: 1-62708-011-2.